

Level 3 Practice Programs

Java program solutions without comments

Each question is shown with a clean, compile-ready solution.

Question 1: Shortest, Tallest, and Mean Height

```
import java.util.Random;

public class FootballTeamHeights {
    public static int sum(int[] heights) {
        int total = 0;
        for (int height : heights) {
            total += height;
        }
        return total;
    }

    public static int shortest(int[] heights) {
        int min = heights[0];
        for (int height : heights) {
            if (height < min) {
                min = height;
            }
        }
        return min;
    }

    public static int tallest(int[] heights) {
        int max = heights[0];
        for (int height : heights) {
            if (height > max) {
                max = height;
            }
        }
        return max;
    }

    public static double mean(int[] heights) {
        return (double) sum(heights) / heights.length;
    }

    public static void main(String[] args) {
        int[] heights = new int[11];
        Random random = new Random();

        for (int i = 0; i < heights.length; i++) {
            heights[i] = random.nextInt(101) + 150;
        }

        System.out.print("Heights: ");
        for (int height : heights) {
            System.out.print(height + " ");
        }
        System.out.println();
        System.out.println("Shortest height: " + shortest(heights));
        System.out.println("Tallest height: " + tallest(heights));
        System.out.println("Mean height: " + mean(heights));
    }
}
```

} }

Question 2: NumberChecker Utility Class

```
import java.util.Scanner;

class NumberChecker {
    public static int countDigits(int number) {
        number = Math.abs(number);
        if (number == 0) {
            return 1;
        }
        int count = 0;
        while (number > 0) {
            count++;
            number /= 10;
        }
        return count;
    }

    public static int[] storeDigits(int number) {
        number = Math.abs(number);
        int count = countDigits(number);
        int[] digits = new int[count];
        for (int i = count - 1; i >= 0; i--) {
            digits[i] = number % 10;
            number /= 10;
        }
        return digits;
    }

    public static boolean isDuckNumber(int[] digits) {
        boolean hasZero = false;
        for (int digit : digits) {
            if (digit == 0) {
                hasZero = true;
                break;
            }
        }
        return hasZero;
    }

    public static boolean isArmstrongNumber(int number, int[] digits) {
        int power = digits.length;
        long sum = 0;
        for (int digit : digits) {
            sum += (long) Math.pow(digit, power);
        }
        return sum == Math.abs(number);
    }

    public static int largest(int[] digits) {
        int max = Integer.MIN_VALUE;
        for (int digit : digits) {
            if (digit > max) {
                max = digit;
            }
        }
        return max;
    }

    public static int secondLargest(int[] digits) {
        int first = Integer.MIN_VALUE;
        int second = Integer.MIN_VALUE;
        for (int digit : digits) {
            if (digit > first) {
                second = first;
                first = digit;
            } else if (digit < first && digit > second) {
                second = digit;
            }
        }
        return second;
    }

    public static int smallest(int[] digits) {
```

```

        int min = Integer.MAX_VALUE;
        for (int digit : digits) {
            if (digit < min) {
                min = digit;
            }
        }
        return min;
    }

    public static int secondSmallest(int[] digits) {
        int first = Integer.MAX_VALUE;
        int second = Integer.MAX_VALUE;
        for (int digit : digits) {
            if (digit < first) {
                second = first;
                first = digit;
            } else if (digit > first && digit < second) {
                second = digit;
            }
        }
        return second;
    }
}

public class NumberCheckerDemo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        int[] digits = NumberChecker.storeDigits(number);

        System.out.println("Count of digits: " + NumberChecker.countDigits(number));

        System.out.print("Digits array: ");
        for (int digit : digits) {
            System.out.print(digit + " ");
        }
        System.out.println();

        System.out.println("Duck number: " + NumberChecker.isDuckNumber(digits));
        System.out.println("Armstrong number: " + NumberChecker.isArmstrongNumber(number, digits));
        System.out.println("Largest digit: " + NumberChecker.largest(digits));
        System.out.println("Second largest digit: " + NumberChecker.secondLargest(digits));
        System.out.println("Smallest digit: " + NumberChecker.smallest(digits));
        System.out.println("Second smallest digit: " + NumberChecker.secondSmallest(digits));
        sc.close();
    }
}

```